

RefDes	Qty	Value	Manufacturer	Manufacturer Part Number
U1	1	Offline Switch	Power Integrations	TNY285PG
U2	1	Shunt Regulator	Texas Instruments	TL431ACLPR
U3	1	Optocoupler	Lite-On	LTV-817S-TA1-D
T1	1	Custom HF Transformer	Hand-wound	N/A
L1	1	1 mH	TDK / EPCOS	B82464A4105K000
F1	1	Fuse	Littelfuse	031300.5MXP
MOV1	1	MOV	Panasonic	ERZ-V14D271
BR1	1	Bridge Rectifier	Vishay	DF04S-E3/77
Cbulk1	1	47 μ F	Panasonic	EEU-EB2G470
Cbulk2	1	47 μ F	Panasonic	EEU-EB2G470
Dclamp	1	Ultrafast Diode	Vishay	UF4007-E3/73
Cclamp	1	2.2 nF	TDK	B32672L8222K000
Rclamp	1	220 k Ω	Yageo	FMP100JR-52-220K
Rseries	1	47 Ω	TE Connectivity	ROX5SSJ47R
DZ1	1	TVS 250 V (Z clamp)	Littelfuse	SMBJ250A
Rstartup	1	1 M Ω	Vishay	MRS25000C1004FCT00
Rbleed	1	820 k Ω	Vishay	MRS25000C8203FCT00
Dsec	1	Schottky Diode	Diodes Inc.	DFLS240-7
Cout1	1	470 μ F	Panasonic	EEU-FC1C471
Lout	1	1 mH	TDK / EPCOS	B82464A4105K000
Rupper	1	15.4 k Ω	Yageo	MFR-25FTE52-15K4
Rlower	1	3.3 k Ω	Vishay	MRS25000C3301FRP00
Ccomp	1	1 nF	Murata	C1206C102J1HACAUTO
RLED	1	1 k Ω	Vishay	MRS25000C1001FCT00

Distributor	Distributor Part Number	Package
Digi-Key	568-13488-ND	DIP-8
Digi-Key	296-17881-1-ND	TO-92
Digi-Key	425-1957-1-ND	SO-4
—	—	Custom
Digi-Key	495-2004-1-ND	SMD
Digi-Key	F4805-ND	3AG
Digi-Key	ERZV14D271-ND	Disc
Digi-Key	78-DF04S-E3/77DICT-ND	DFS
Digi-Key	P13677-ND	Radial
Digi-Key	P13677-ND	Radial
Digi-Key	2142323	DO-41
Digi-Key	495-2009-1-ND	Film
Digi-Key	2058591	Axial
Digi-Key	10231882	Axial
Digi-Key	SMBJ250ABBCT-ND	DO-214AA (SMBJ)
Digi-Key	BC1M0-ND	Axial
Digi-Key	BC820K-ND	Axial
Digi-Key	DFLS240-7DICT-ND	PowerDI123
Digi-Key	P10248-ND	Radial
Digi-Key	495-2004-1-ND	SMD
Digi-Key	13-MFR-25FTE52-15K4CT-ND	Axial
Digi-Key	PPC3.30KZCT-ND	Axial
Digi-Key	399-C1206C102J1HACAUTOCT-ND	1206
Digi-Key	BC3325CT-ND	Axial

Rating	Function
700 V MOSFET	Flyback controller
2.495 V ref	Error amplifier
CTR opto	Isolated feedback
—	Isolation + energy transfer
330 mA	Input π filter choke
0.5 A 250 VAC	Input protection
175 VAC	Surge protection
1 A / 400 V	AC to DC
400 V	Bulk capacitor
400 V	Bulk capacitor
1 kV / 1 A	RCD clamp diode
2 kV	Clamp capacitor
1 W	Clamp resistor
5 W	Drain damping
250 V standoff, 405 V clamp	Z clamp (RCD+Z backstop)
0.6 W	EN/UV startup
0.6 W	Bulk bleeder
40 V / 2 A	Output rectifier
16 V	Output bulk
330 mA	Output π filter inductor
0.25 W	TL431 divider top
0.6 W	TL431 divider bottom
100 V	Loop compensation
0.6 W	Opto LED resistor

Notes
Primary IC
Feedback loop
Opto LED via RLED
Designed & wound in-house
Between bulk caps
Slow-blow
Across AC line
Mains rectifier
Pre-choke
Post-choke
Ultrafast
Leakage energy
Sets dissipation
Series with clamp
Backstop clamp across primary; optional but included as 'Z'
HV bias
Discharge
5 V rail
Main filter
Same part as input choke; used on output
Sets Vout
Sets Vout
TL431 cathode
CTR bias

Target Vout	Rtop (Vout→REF)	Rbot (REF→GND)	Divider Current @ Vout
≈5.0 V	2.21 kΩ	2.21 kΩ	~1.13 mA
≈5.0 V	1.0 kΩ	1.0 kΩ	~2.5 mA
≈5.0 V	4.7 kΩ	4.7 kΩ	~0.53 mA
≈12 V	8.66 kΩ	2.21 kΩ	~1.1 mA
≈20 V	15.4 kΩ	2.21 kΩ	~1.2 mA

Resulting Vout	Notes
≈4.99 V	Equal resistors; ideal first bring-up
≈4.99 V	Stronger loop, more dissipation
≈4.99 V	Still OK for TL431A/B
≈12.0 V	Scaled from 5 V divider
≈20.0 V	Good for future UCC flyback